Case Study: Spread of COVID-19 in Three Semantics

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Case & model. SEIRD epidemic in a closed, well-mixed population: $N = 100$, one index case, $R_0 = 2.5$, latent period 5.2 d, infectious period 7 d, fatality 1 %. Built from scratch three times: stochastic timed automata (one `Person` per individual), a CTMC (four `Reaction` processes racing with rates `propensity(e)`), and ODEs (clocks `Sus`, ..., `Dead` with derivatives in an invariant).

User & goal. A user asked the assistant (Claude, connected to UPPAAL via MCP) to "create a model of spreading of COVID in a population of a given size", then to refactor it as a CTMC and as an ODE model.

Interaction. Agent-based model → CTMC → ODE → stale engine detected → "why the clock?" → clock removed → CTMC re-run.

Table 1. SMC estimates per model variant (95 % CI; 500 runs for expectations; –: not applicable, the ODE cannot die out). Valid: computed on the model shown in the editor.

Variant	procs	valid	peak `Inf`	deaths	≥ 50 % infected	early extinction
Agent-based	100	✓	11.61 ± 0.83	0.53 ± 0.07	0.56 ± 0.05	0.40 ± 0.05
CTMC, first run	4	× (agent)	11.54 ± 0.82	0.45 ± 0.07	0.61 ± 0.05	0.37 ± 0.05
CTMC, re-run	4	✓	11.43 ± 0.82	0.52 ± 0.07	0.58 ± 0.05	0.41 ± 0.05
ODE + sampling clock	1	✓	13.386	0.8937	final size 89.37	–
ODE	1	✓	13.389	0.8938	final size 89.38	–

Key findings.

F1. The engine verified a model other than the one in the editor. `replaceModel` does not reload the verifier; only targeted edits do. 21 results came from the agent-based model and looked right for the CTMC, since both describe the same Markov chain. The deterministic ODE giving run-dependent integers exposed it; probe queries on identifiers unique to the intended model (`h`, `t`, `INFECTION+DEATH`) became the reload check.

F2. Three semantics agreed with each other and with theory. Agent-based and CTMC intervals overlap in every row; early extinction $\approx 1/R_0 = 0.40$; ODE final size matches $z = 1 - e^{-R_0 z} \approx 0.893$; the ODE peak is higher because stochastic means include early die-outs.

F3. The assistant's explanations were the weak link. Two diagnoses of the stale engine were wrong: renaming variables to avoid the query keyword `E`, and adding a 0.1-day sampling clock. The user's question removed the clock; results changed in the 4th digit.

User's perspective. Asking for the same epidemic in another semantics took one sentence and gave an independent check; a "why is this here?" question caught an unneeded construct; "re-run properly" turned a suspicion into a clean result.

Tool & interface lessons. (i) Whole-model replacement leaves the verifier on the old model. (ii) Results carry no fingerprint of the model they were computed on. (iii) Models live only in the editor; nothing is saved to disk unless asked.

Table 2. UPPAAL MCP tool calls in the session.

Purpose	Tools (calls)	Total
Inspect	`getContext` (2), `getModel` (3)	5
Verify	`verifyQueries` (15)	15
Queries	`insertQuery` (7), `removeQuery` (5)	12
Edit	`replaceModel` (6), `checkSyntax` (6), `modifyLocation` (2), `modifyEdge` (2), `setGlobalDeclarations` (2), `deleteEdge` (1)	19
All	11 distinct tools	51

Metrics. 6 prompts; 51 tool calls, 0 failed; 15 verification runs (60 query checks: 21 on a stale model, 8 with query errors, 4 probes); 3 semantics, 5 variants; 2 wrong diagnoses corrected.